DP Biology – B4.1

Adaptation to environment

- How do a species' adaptations relate to its habitat?
- Why do ecosystems in the same land biome resemble each other?

SL statements B4.1.1–8. Textbook pp. 341–362.
Image: marram grass on a sand dune. W.carter, CC0.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	B4.1.1–2	Describe a habitat, and explain how plants are adapted to sand dunes and mangrove swamps
2	B4.1.3–5	Explain how abiotic variables affect distribution, use transect data to find a range of tolerance, and state the conditions for coral reefs
3	B4.1.6–8	Use climate to explain biomes, explain why biomes have similar communities, and describe adaptations to deserts and rainforests

The big picture: fitting the place

Habitat	Tolerance	Biome
Where it lives	Which conditions it survives	Climate decides the community

Abiotic conditions such as water, salt, light and temperature select **adaptations**. Similar climates produce similar communities worldwide.

Big question

Why do cacti in America and euphorbias in Africa look almost the same?

Habitat

B4.1.1

A **habitat** is where an organism, population or community lives: geographical location, physical location and ecosystem type.

A **microhabitat** is a small spot within it, such as under a log.

Example

African savanna elephant: grassland and open woodland in sub-Saharan Africa.

Activity

Describe the habitat of a species you know, at all three levels.

Adaptation to the abiotic environment

B4.1.2

Biotic: living

predators, food, competitors, parasites

Abiotic: non-living

temperature, water, light, pH, salinity

Alpine plants

Edelweiss: white woolly hairs reflect UV and trap warm air; low cushion growth avoids wind. Each feature is a heritable **adaptation** to abiotic stress.

Grass on sand dunes

B4.1.2



Marram grass. W.carter, CC0.

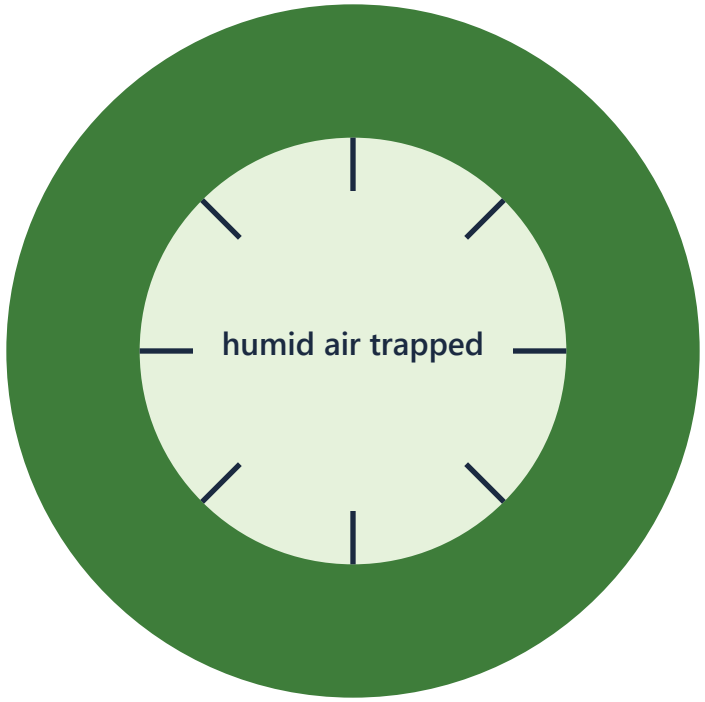
Feature	How it helps
Rolled leaves	Trap moist air
Stomata in pits	Less transpiration
Thick waxy cuticle	Less water loss
Deep rhizomes	Reach water; hold sand

The challenge

Sand holds little water or minerals; salt spray and wind increase water loss.

Inside a rolled marram leaf

B4.1.2



thick waxy outer cuticle

hairs and sunken stomata inside

Key idea

Water vapour builds up inside the roll, so the gradient for water to leave is smaller.

Trees in mangrove swamps

B4.1.2



Pneumatophores. Katja Schulz, CC BY 2.0.



Red mangrove roots. S. Hillebrand, USFWS, public domain.

Feature	How it helps
Pneumatophores	Roots grow up into air: oxygen
Stilt roots	Support in soft mud
Salt glands	Remove excess salt
Thick waxy leaves	Less water loss

The challenge

Waterlogged, anaerobic, salty mud that moves with the tide.

Practice: adapted plants

B4.1.1–2

Explain

Two marram adaptations to drought.

Explain

Why mangrove roots grow upwards.

Distinguish

Habitat and adaptation.

Use

xerophyte, transpiration, anaerobic, salinity

Check-in: class 1

B4.1.1–2

1. Define habitat.
2. Distinguish biotic and abiotic factors.
3. Give two marram adaptations.
4. Give two mangrove adaptations.
5. Explain how rolled leaves reduce water loss.

How to use this

Answer without your notes, then check against slides 3–9.

Abiotic variables and distribution

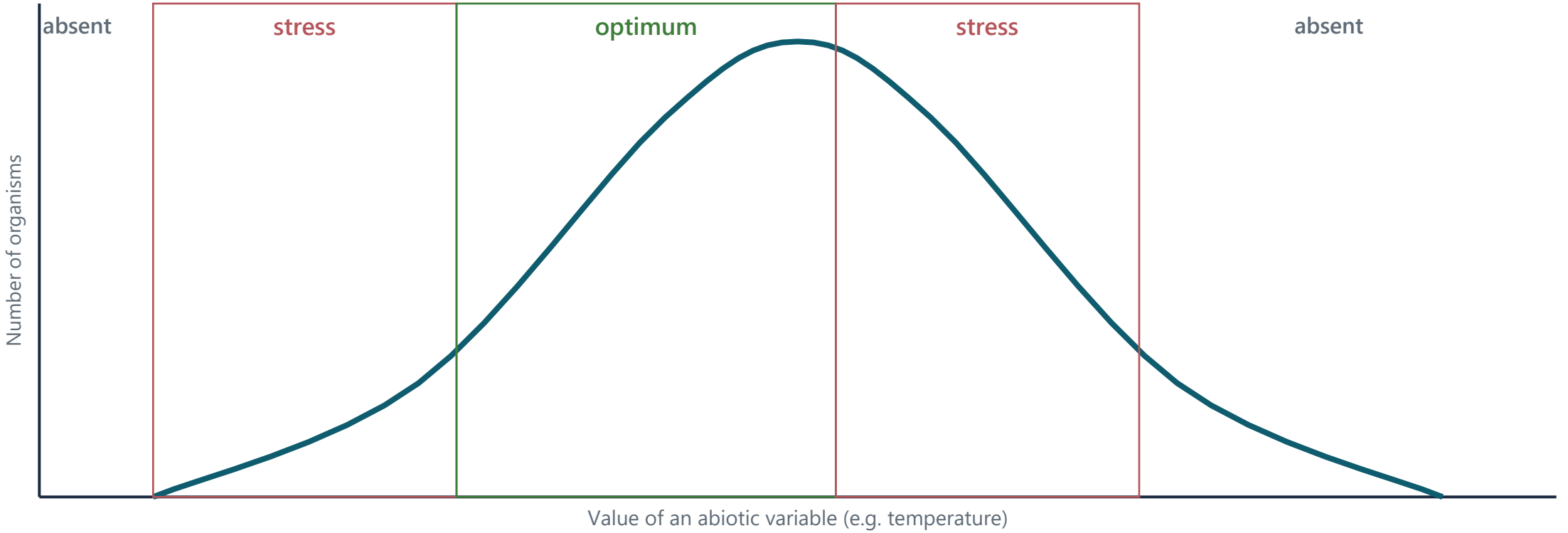
B4.1.3

Group	Abiotic variables
Plants	Temperature, water, light, soil pH, salinity, minerals
Animals	Temperature, water, breeding sites, food, territory

Key idea

A species has a **range of tolerance** for each variable: inside it, survival and reproduction; near the limits, stress; beyond them, death.

Range of tolerance



Limiting factors

B4.1.4

The **limiting factor** restricts where a species grows: light on a shaded slope, salinity on a salty shore.

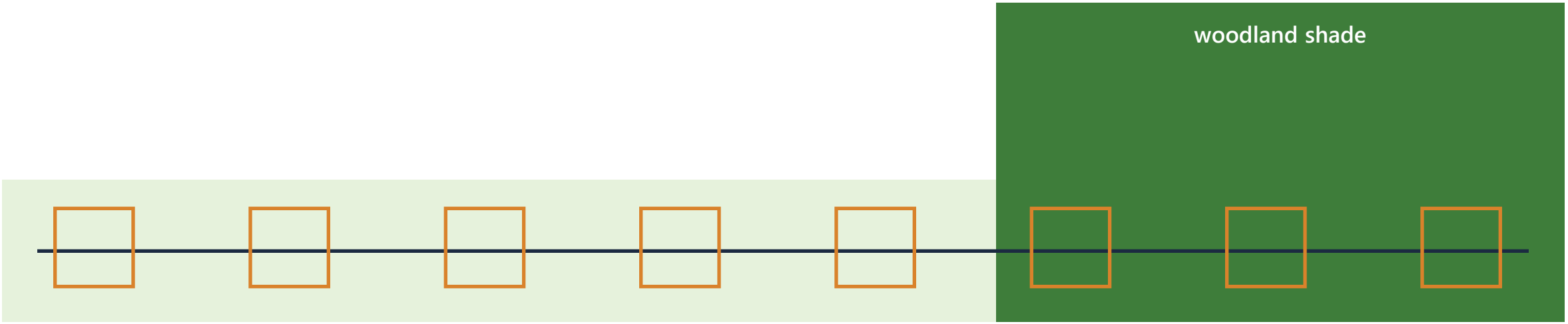
A species may tolerate one factor widely and another narrowly.

Think

A plant grows at the top of a salt marsh but not the bottom. Name a likely limiting factor at the bottom, and suggest a different one at the top.

Using a transect

B4.1.4



tape line with quadrats every 10 m

1. Record species and numbers in each quadrat
2. Measure the abiotic variable (light) at each point
3. Plot both against distance; look for a correlation

Working with data: a salt-marsh transect

B4.1.4

Plant cover and soil salinity along a transect from the sea. Invented values for practice.

Distance / m	Salinity / g dm^{-3}	Cover / %
0	34	0
20	28	15
40	20	60
60	12	75
80	5	30

1. Describe the change in cover.
2. Why is cover zero at 0 m?
3. Why does cover fall at 80 m?

Practice: transects and tolerance

B4.1.3–4

Explain	Sketch	Design
Why correlation does not prove cause.	A tolerance curve, labelled.	A transect into woodland.

Practical

Collect your own transect data in a natural area and identify one plant’s range of tolerance.

Coral reefs: animals with partners

B4.1.5



Coral reef. Jim Maragos, public domain.

Reef corals are **animals** hosting photosynthetic algae (**zooxanthellae**) in their tissues.

So reefs need conditions where the algae can photosynthesize.

Mutualism

Algae supply sugars; corals supply shelter, CO_2 and nutrients.

Conditions for coral reefs

B4.1.5

Factor	Condition	Why
Depth	Mostly under 25 m	Light reaches algae
pH	Above 7.8	Calcium carbonate can form
Salinity	About 35 parts per thousand	Avoids osmotic stress
Clarity	Clear water	Light passes through
Temperature	About 23–29 °C	Suits corals and algae

Threats

Warming causes bleaching; ocean acidification lowers pH.

Practice: reef or no reef?

Site	Depth / m	pH	Temp / °C	Water
A	10	8.1	27	Clear
B	60	8.1	26	Clear
C	8	8.1	26	Muddy river outflow
D	12	8.1	16	Clear

Task

Which site could support a reef? Give the limiting factor at each other site.

Check-in: class 2

B4.1.3–5

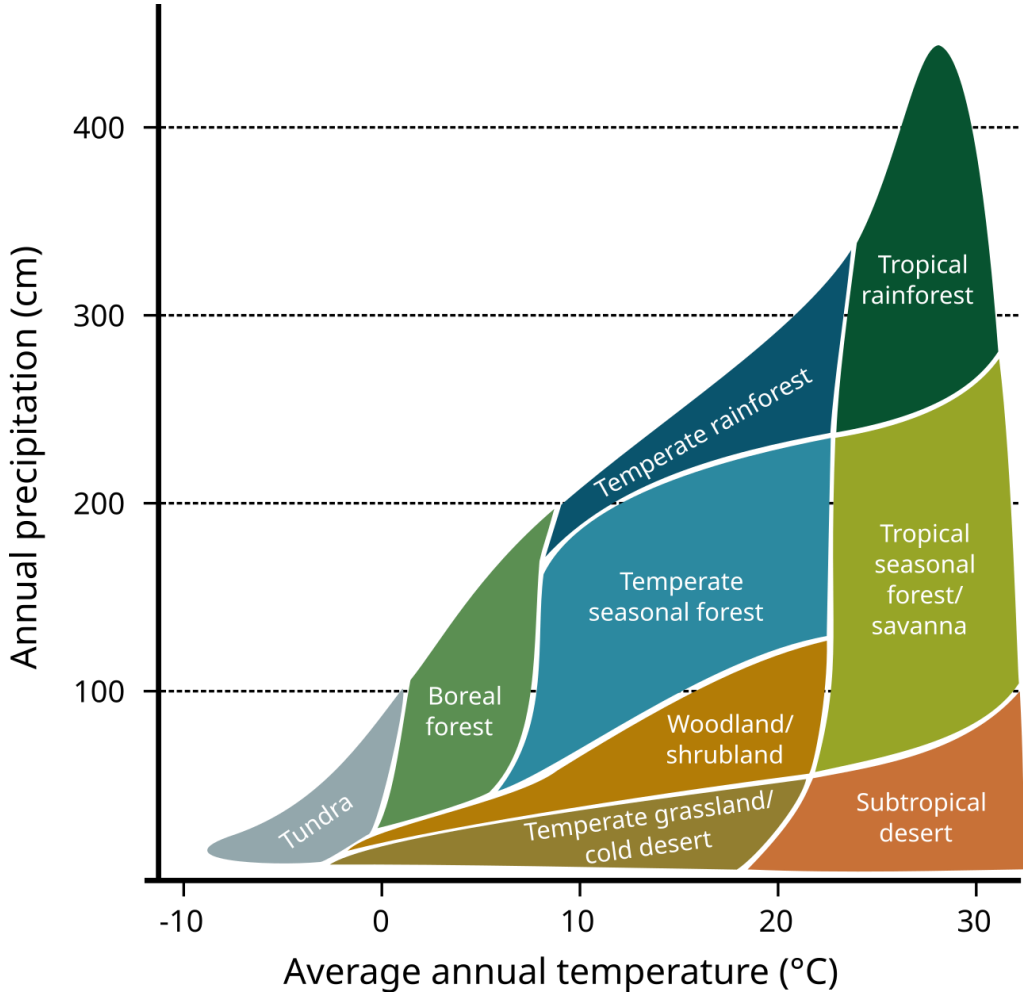
1. Define range of tolerance.
2. Define limiting factor.
3. Describe how to carry out a transect.
4. State four conditions coral reefs need.
5. Explain why reef corals need light.

How to use this

Answer without your notes, then check against slides 11–19.

Climate determines land biomes

B4.1.6



Biomes by temperature and rainfall. Navarras, CC0.

Temperature and **rainfall** decide which natural ecosystem develops.

So a climate predicts the biome.

Activity

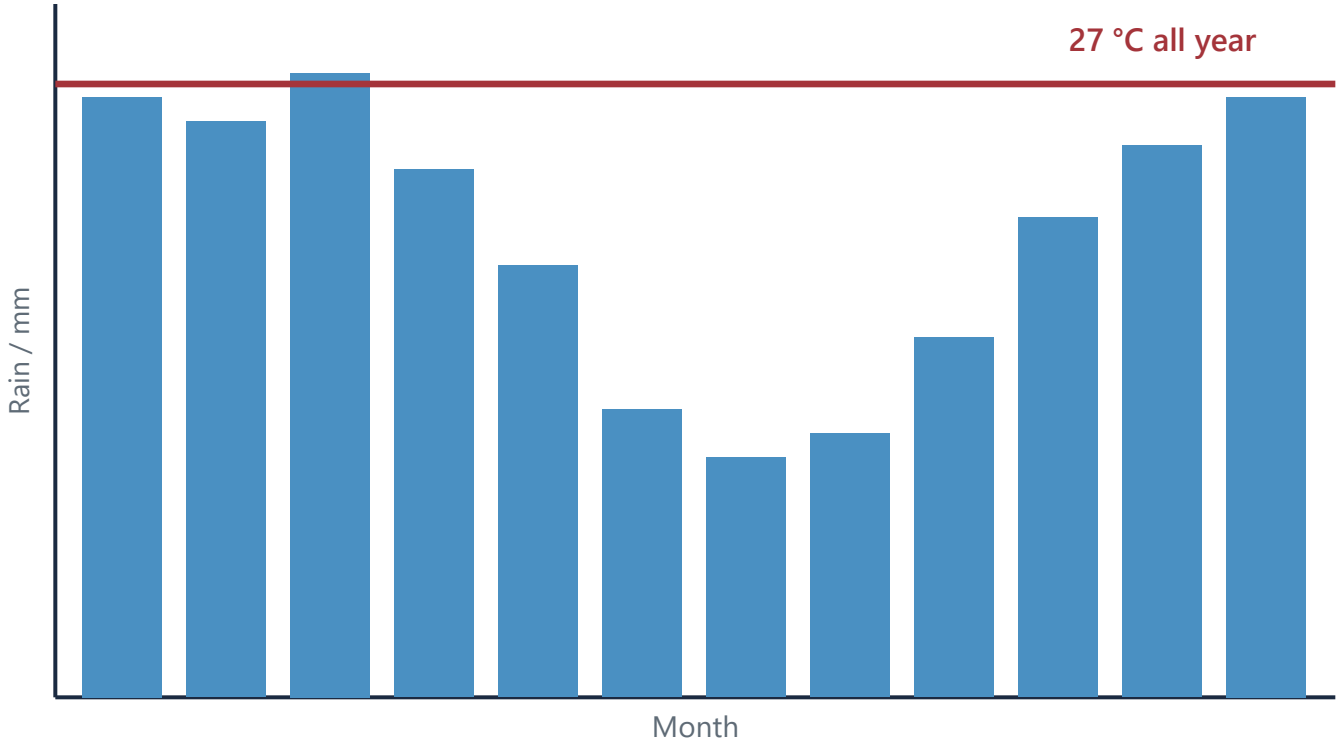
Mean 25 °C, 300 cm (3000 mm) of rain per year: which biome?

Six biomes

B4.1.7

Biome	Temperature	Rainfall	Plants
Tropical forest	High all year	High all year	Very diverse
Temperate forest	Warm summer, cold winter	Moderate	Deciduous trees
Taiga	Long cold winters	Low to moderate	Conifers
Grassland	Seasonal	Low to moderate	Grasses
Tundra	Very cold	Low	Low plants, no trees
Hot desert	Very hot days	Very low	Sparse, water stores

Reading a climate graph



Which biome?

Hot all year, wet all year: tropical rainforest.

Why biomes look alike around the world

B4.1.7



Tundra in Alaska. USFWS, public domain.



Siberian taiga. V. Argenberg, CC BY 4.0.

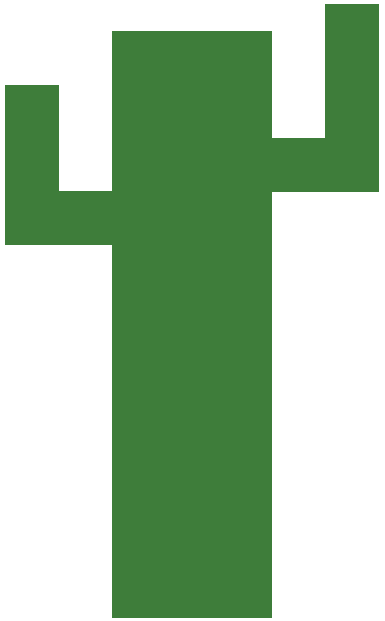
Similar conditions give similar **selection pressures**.

Key idea

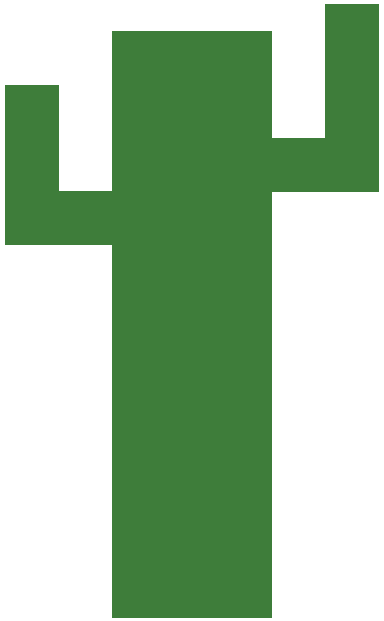
Tundra plants in Canada and Russia are often not closely related, yet all grow low and slowly. Similar climate + convergent evolution = similar communities.

Convergent evolution in deserts

B4.1.7–8



American cactus



African euphorbia

Big question

Unrelated plants, same climate, same shape: thick water-storing stems, no leaves, spines.

Adaptations to hot deserts

B4.1.8



Saguaro. Angwinar, CC0.



Fennec fox. Drew Avery, CC BY 2.0.

Species	Adaptations
Saguaro	Water-storing stem; spines; wide shallow roots
Fennec fox	Large ears lose heat; nocturnal; furry paws

The challenge

Hot days, cold nights, little water, thin soil.

Adaptations to tropical rainforests

B4.1.8



Spider monkey. 25 OClock, CC0.



Monstera. Cliff, CC BY 2.0.

Species	Adaptations
Spider monkey	Long limbs; gripping tail
Poison dart frog	Toxins; warning colours
Climbers	Climb to light; drip tips



Blue poison dart frog. A. Pingstone, public domain.

The challenge

Hot, wet, dark under the canopy; fierce competition and predation.

Check-in: class 3

B4.1.6–8

1. Name the two climate factors that decide biomes.
2. Describe tundra climate and plants.
3. Explain why biomes look alike worldwide.
4. Give two desert adaptations.
5. Give two rainforest adaptations.

How to use this

Answer without your notes, then check against slides 21–27.

Common mistakes to avoid

Not this	But this
Organisms adapt because they need to.	Adaptations arise by natural selection.
A biome is one ecosystem.	A biome is a group of ecosystems with similar climate.
Reefs need nutrient-rich water.	They need clear, low-nutrient water.
A transect correlation proves cause.	Another factor may vary too.

Summary: adaptation to environment

B4.1.1–8

	Key idea
B4.1.1	Habitat
B4.1.2	Dune grasses and mangroves
B4.1.3–4	Abiotic variables; range of tolerance; transects
B4.1.5	Coral reef conditions
B4.1.6	Climate decides biomes
B4.1.7	Convergent communities
B4.1.8	Desert and rainforest adaptations

Check that you can...

1. **Distinguish** habitat and adaptation
2. **Explain** dune and mangrove adaptations
3. **Explain** tolerance and limiting factors
4. **Outline** a transect
5. **Explain** how climate determines biomes
6. **Explain** convergent evolution in biomes

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State two abiotic variables that affect the distribution of plants.
2. Explain how pneumatophores help a mangrove tree.
3. Explain why tundra plants on different continents look alike.

Image credits

Marram grass on dune at Nørre Vorupør Strand.jpg – W.carter – CC0

Black Mangrove Pneumatophores – Flickr – treegrow.jpg – Katja Schulz from Washington, D. C., USA – CC BY 2.0

Close view of red mangrove roots.jpg – Steve Hillebrand, U.S. Fish and Wildlife Service – Public domain

Coral reef fish pacific blue tan paracanthurus hepatus.jpg – Jim Maragos, U.S. Fish and Wildlife Service – Public domain

Climate influence on terrestrial biome.svg – Navarras – CC0

Tundra landscape (6763177589).jpg – USFWSAlaska – Public domain

Lake Baikal, Pine forest, Siberian taiga, Siberia, Russia.jpg – Vyacheslav Argenberg – CC BY 4.0

Saguaro Cactus In Arizona, Sonoran Desert-2013.jpg – Angwinar – CC0

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